

Section 2

Solutions of Equations in One Variable (Root-Finding)

Root-finding

Definition

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ (univariate), then x is called a **root**, or **zero**, of f if $f(x) = 0$.

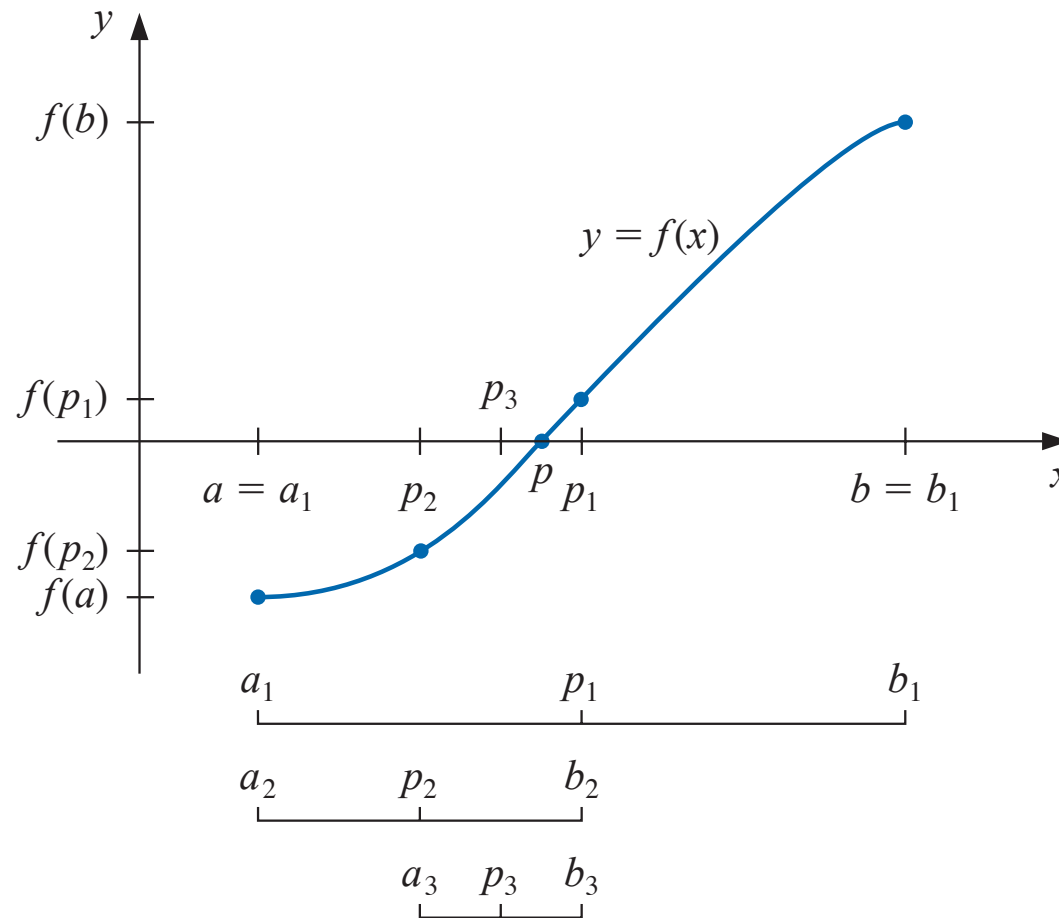
Example (Roots of a function)

Find the root(s) of $f(x)$ defined by (a) $(x - 1)(x + 1)$; (b) $(x - 1)^2$; (c) $x^2 + 1$; (d) $ax^2 + bx + c$; (e) $\cos(x)$.

Question: Given a general function f , how can we find its root/roots?

Bisection method

Suppose f is *continuous* on $[a, b]$, and $f(a)f(b) < 0$ (WLOG $f(a) < 0, f(b) > 0$). Then f has at least one root in (a, b) by IVT.



Bisection method

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Bisection method

- ▶ **Input.** Endpoints a, b . Tolerance ϵ_{tol} . Maximum number of iterations N_{max} . Set iteration counter $N = 1$.
- ▶ While $N \leq N_{\text{max}}$, do
 1. Set $p = \frac{a+b}{2}$, compute $f(p)$. If $f(p) = 0$ or $b - a < \epsilon_{\text{tol}}$, break.
 2. If $f(p) > 0$, set $b = p$. If $f(p) < 0$, set $a = p$.
 3. $N \leftarrow N + 1$.
- ▶ **Output.** If $i = N_{\text{max}}$, print(“Maximum iteration reached.”). Return p .

Termination condition

Bisection method can run forever if we do not set termination condition (e.g., ϵ_{tol} , N_{max}).

Common choices of termination condition:

- ▶ Fixed number of iterations N_{max} .
- ▶ $|p_N - p_{N-1}| < \epsilon_{\text{tol}}$
- ▶ $|f(p_N)| < \epsilon_{\text{tol}}$
- ▶ $\frac{|p_N - p_{N-1}|}{|p_N|} < \epsilon_{\text{tol}}$

Example

Example (Bisection method)

$f(x) = x^3 + 4x^2 - 10$. Find a root in $[1, 2]$ using the bisection method.

Solution. First check if $f(1)f(2) < 0$ (if not, bisection method may not apply).

Bisection method

Theorem

Suppose $f \in C[a, b]$ and $f(a)f(b) < 0$, then p_n generated by the bisection method converges to p , a root of f , with $|p_n - p| < \frac{b-a}{2^n}$

Drawbacks of the bisection method:

- ▶ inefficient
- ▶ may discard some roots